

# Cybersecurity Incident Report:

## Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

As part of the DNS protocol, the UDP protocol contacts the DNS server to retrieve the IP address for the domain name of yummyrecipesforme.com. The ICMP protocol responded with an error message, indicating the issue with contacting the DNS server.

The UDP message going from your browser to the DNS server is shown in the first two lines of every log event. The ICMP error response from the DNS server to your browser is displayed in the third and fourth lines of every log event with the error message, “udp port 53 unreachable.” Since port 53 is associated with DNS protocol traffic, we know this is an issue with the DNS server. Issues with performing the DNS protocol are further evident because the plus sign after the query identification number 35084 indicates flags with the UDP message, and the “A?” symbol indicates flags with performing DNS protocol operations

Due to the ICMP error messages about port 53, the most likely issue is with the DNS server not responding. This assumption is further supported by the flags associated with the outgoing UDP message and domain name retrieval. This may indicate a DoS or DDoS attack on the DNS server, where its services are being overwhelmed.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The incident first occurred earlier in the day and was later verified at 1:24 pm.

The IT team became aware of the incident through customer complaints to the organization, reporting that they received the message “destination port unreachable” when they attempted to visit the website yummyrecipesforme.com.

The actions of the cybersecurity team providing IT services were to first verify the issue by attempting to visit the website, then to troubleshoot and run tests on the issue, using a network analyzer tool, tcodump, to conduct packet sniffing tests.

The resulting logs revealed that port 53, used for connections to the DNS server, was unreachable. The next step is to identify whether the DNS server is down or if traffic to

port 53 is blocked by the firewall.

A likely cause of the incident would be a DoS or DDoS attack, overflowing the DNS server with requests, causing it to shut down or a misconfiguration.